

NUMERICAL REALIZATION OF AN ENTROPY-BASED SELECTION PRINCIPLE IN THE EXAMPLE OF KELVIN-HELMHOLTZ INSTABILITY*

SHAOSHUAI CHU[†] AND MICHAEL HERTY[‡]

Abstract. In this paper, we extend numerical schemes based on parameterized Young measures and linear programming to two space dimensions and apply them to the Kelvin-Helmholtz instability governed by the two-dimensional Euler equations of gas dynamics. We construct first-, second-, third-, fifth-, seventh-, and ninth-order Young-measure schemes and compare them with corresponding standard local Lax-Friedrichs (LLF) flux-splitting schemes and LLF schemes equipped with an entropy-based local characteristic decomposition (LLF-ELCD). We examine instantaneous and time-averaged density profiles, cumulative averages across spatial reconstruction orders, regional empirical density distributions, averaged density marginals, and several selection criteria for dissipative weak solutions. The numerical results reveal scheme-dependent flow patterns, particularly in the small-scale structures generated during the roll-up of the shear layers. At every reconstruction order considered, the Young-measure schemes yield the largest time-averaged physical entropy. By contrast, the LLF-ELCD schemes do not systematically yield larger entropy than the standard LLF schemes and therefore do not constitute a consistent maximum-entropy selection mechanism. The averaged density marginals of the Young measures are concentrated on several neighboring density states, and the later cumulative averages are closer for all Young-measure schemes. These results demonstrate the feasibility of the two-dimensional Young-measure formulation and show that the objective function in the linear-programming problem can act as an effective selection mechanism. The observed entropy preference is consistent with the local optimization of the expected physical entropy. It cannot be reproduced by merely incorporating entropy into a conventional numerical construction.

Key words. Dissipative weak solutions; entropy-based selection principle; Euler equations of gas dynamics; high-order schemes; Kelvin-Helmholtz instability; Young measures.

MSC codes. 65M08, 65M12, 76M12, 76M20, 76N10, 35L65.

1. Introduction. In this paper, we consider the two-dimensional (2-D) Euler equations of gas dynamics:

$$(1.1) \quad \begin{aligned} \rho_t + (\rho u)_x + (\rho v)_y &= 0, \\ (\rho u)_t + (\rho u^2 + p)_x + (\rho uv)_y &= 0, \\ (\rho v)_t + (\rho uv)_x + (\rho v^2 + p)_y &= 0, \\ E_t + [u(E + p)]_x + [v(E + p)]_y &= 0. \end{aligned}$$

Here, x and y are the spatial variables, t is time, ρ is the density, u and v are the velocities in the x - and y -directions, respectively, E is the total energy density, and p is the pressure. The system (1.1) is closed by the polytropic equation of state

$$(1.2) \quad e(\rho, \theta) = c_v \theta, \quad c_v = \frac{1}{\gamma - 1}, \quad p(\rho, \theta) = \rho \theta = (\gamma - 1) \rho e = (\gamma - 1) \left(E - \frac{|\mathbf{m}|^2}{2\rho} \right),$$

where e is the specific internal energy, θ is the absolute temperature, $\mathbf{m} = (\rho u, \rho v)^\top$ is the momentum vector, and $\gamma > 1$ is the ratio of specific heats. The second law of thermodynamics is expressed through

$$(1.3) \quad S_t + (Su)_x + (Sv)_y \geq 0, \quad S := c_v \rho \ln \left(\frac{p}{\rho^\gamma} \right),$$

where S is the physical entropy density. Correspondingly, $\eta := -S$ is the mathematical entropy, for which the inequality in (1.3) is reversed. On a bounded domain $\Omega \subset \mathbb{R}^2$ with periodic or no-flux boundary conditions, we prescribe

$$(1.4) \quad \rho(x, y, 0) = \rho_0(x, y), \quad \mathbf{m}(x, y, 0) = \mathbf{m}_0(x, y), \quad E(x, y, 0) = E_0(x, y).$$

*Submitted to the editors DATE???

Funding: The work of S. Chu and M. Herty was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – SPP 2410 Hyperbolic Balance Laws in Fluid Mechanics: Complexity, Scales, Randomness (CoScaRa) within the Project HE5386/27-2 (Zufällige kompressible Euler Gleichungen: Numerik und ihre Analysis, 525853336).

[†]Department of Mathematics, RWTH Aachen University, Aachen, 52056, Germany (chu@igpm.rwth-aachen.de).

[‡]Department of Mathematics, RWTH Aachen University, 52056 Aachen, Germany; Department of Mathematics and Applied Mathematics, University of Pretoria, Hatfield, 0028, South Africa (herty@igpm.rwth-aachen.de).

The corresponding initial pressure and physical entropy are

$$p_0 = (\gamma - 1) \left(E_0 - \frac{|\mathbf{m}_0|^2}{2\rho_0} \right), \quad S_0 = c_v \rho_0 \ln \left(\frac{p_0}{\rho_0^\gamma} \right).$$

Solutions of (1.1)–(1.2) may develop shocks, rarefaction waves, and contact discontinuities even from smooth initial data and must then be understood in the weak sense. In one space dimension, the entropy inequality provides an important admissibility criterion. In several space dimensions, however, entropy admissibility alone does not guarantee uniqueness. Convex-integration results show that multidimensional compressible Euler systems may admit infinitely many admissible weak solutions for certain initial data; see, e.g., [4, 10, 12] and the references therein.

This lack of a general uniqueness theory makes the interpretation of numerical approximations particularly delicate for multidimensional shear flows. As reported in, e.g., [1, 2, 7, 13, 18], numerical approximations to Kelvin-Helmholtz (KH)-type flows frequently fail to converge strongly under mesh refinement and instead develop increasingly fine vortical structures and oscillations. This raises the fundamental question of how the limiting behavior of such approximations should be characterized and motivates generalized solution concepts capable of describing the resulting oscillations, concentrations, and dissipation. Measure-valued solutions, introduced in [11], provide a natural framework for this purpose: rather than assigning a single state to each space-time point, they assign a parameterized probability measure that records the local distribution of possible states. Measure-valued formulations and their numerical approximation for multidimensional hyperbolic conservation laws were studied in [18] and the references therein. Within this broader framework, we focus on consistent numerical approximations and their relation to dissipative weak (DW) solutions of the Euler equations; see, e.g., [16]. When a sequence of consistent numerical approximations converges only weakly, nonlinear quantities such as the physical fluxes cannot be identified from the weak limit of the conservative variables alone. Consequently, that weak limit need not satisfy the Euler equations in the sense of distributions. In [16, 26], Cesàro averages of consistent approximations computed on successively refined meshes were studied for low-order methods and shown, along suitable subsequences, to converge strongly to a DW solution. Related results were reported in [13]. More recently, a fifth-order alternative weighted essentially non-oscillatory (A-WENO) scheme was used in [3] for multidimensional Euler flows.

Whereas the preceding studies recover meaningful limiting behavior through Cesàro averaging over mesh-refined sequences, we pursue a complementary approach by incorporating parameterized Young measures directly into the numerical formulation. More precisely, we extend the linear-programming (LP)-based Young-measure schemes of [5, 8] from one to two space dimensions and apply them to the KH instability for the compressible Euler equations. At each spatial degree of freedom, the numerical state is represented by a discrete Young measure whose marginal first moments coincide with the corresponding components of the evolved conservative state. Among the measures satisfying the normalization and first-moment constraints, one is selected by solving an LP problem. The measure-valued physical fluxes are then obtained by averaging the Euler fluxes with respect to the selected measure. Because these constraints generally do not determine a unique measure, the objective function plays a central role in the proposed framework. Related comparisons of scheme-dependent DW solutions using cumulative averages, approximate Young measures, entropy production, energy-defect criteria, and a Bregman-functional criterion were carried out in [7]. There, the criteria were applied **a posteriori** to compare already computed solutions. Here, by contrast, the prescribed criterion enters the numerical evolution through the objective function of the LP problem. We use the mathematical entropy $\eta = -S$ as the objective. Minimizing its expectation is equivalent to maximizing the expected physical entropy among the admissible discrete measures. The resulting closure, therefore, selects, for each prescribed moment, a measure with maximum expected physical entropy.

To determine whether this explicit optimization produces an observable entropy preference, we compare three families of schemes: the proposed Young-measure schemes with entropy-based LP closure, standard local Lax–Friedrichs (LLF) flux-splitting schemes, and LLF schemes incorporating the entropy-based local characteristic decomposition (ELCD) of [6]. For each family, we construct first-, second-, third-, fifth-, seventh-, and ninth-order realizations. This allows comparisons between methods of the same formal order. Since the first-order method requires no LCD, the standard LLF and LLF–ELCD schemes coincide at this order. The third family is included, in particular, to determine whether incorpo-

rating entropy into a classical LCD is sufficient to produce an entropy-selective numerical evolution.

The numerical experiments show that the three families of schemes may produce different evolutions of the KH instability, with the clearest differences appearing in the small-scale vortical structures. The density profiles produced by standard LLF and LLF–ELCD are quite similar, whereas noticeable differences arise between these profiles and the Young-measure results, especially at higher orders. The finite averages over different formal orders on a fixed mesh. The time-averaged density profiles are more stable as additional orders are included. These cross-order averages should not be identified with the Cesàro averages discussed above, which are formed by a sequence of consistent approximations as the discretization is refined. The three families produce remarkably similar time-averaged density profiles, while their spatial empirical density distributions retain scheme-dependent differences. These observations provide evidence of numerical selection but neither establish convergence to distinct DW solutions nor rule out strong convergence.

At every formal order considered, the Young-measure scheme yields the largest time-averaged domain-integrated physical entropy among the corresponding schemes. This systematic preference is consistent with the imposed objective and provides numerical evidence that the LP closure affects the selected evolution. By contrast, the LLF–ELCD schemes do not systematically yield larger entropy than standard LLF. Thus, using entropy in the LCD alone does not enforce a maximum-entropy evolution and does not constitute a consistent maximum-entropy selection mechanism. This comparison indicates that the systematic entropy preference of the Young-measure schemes is associated with the explicit optimization performed at each (spatial) degree of freedom. It is not guaranteed by using entropy only as an intermediate component of a conventional method. Moreover, for the averages over all six orders, the Young-measure schemes yield the largest time-averaged total physical entropy, the smallest time-averaged energy defect, and the smallest time-averaged Bregman functional among the three scheme families. By contrast, the LLF–ELCD schemes yield the largest time-averaged energy defect. Therefore, the numerical approximation selected from the three scheme families depends on the prescribed selection criterion.

These findings indicate that the objective function is an active modeling and selection component rather than an auxiliary closure tool. This interpretation points to a broader use of the framework. Although we focus on mathematical entropy, the formulation is not intrinsically restricted to this choice; other objective functions may be introduced to encode alternative physically or mathematically motivated criteria while maintaining the admissibility constraints. The presented framework thus provides a flexible and systematic mechanism for numerical approximations consistent with a prescribed criterion. Optimizing a selected quantity, however, does not by itself imply greater numerical accuracy. In the present experiments, the larger physical entropy shows that the Young-measure approximations are more strongly aligned with the maximum-entropy criterion imposed through the LP.

The rest of the paper is organized as follows. §2 reviews consistent approximations, DW solutions, and the relevant convergence results. §3 presents the 2-D Young-measure, LLF, and LLF–ELCD schemes. §4 reports the KH experiments and compares the resulting structures, entropy evolution, averages, empirical density distributions, and additional selection criteria.

2. Consistent Approximations and Dissipative Weak Solutions. To analyze the convergence of a numerical method, it is essential to investigate its local consistency errors. For this purpose, the notion of a consistent approximation is particularly useful; see, e.g., [16].

For later use, we express the pressure, the specific internal energy, and the total energy density in terms of $(\rho, \mathbf{m}, S)$. For $\rho > 0$, the entropy relation (1.3) yields

$$\begin{aligned} p(\rho, S) &= \rho^\gamma \exp\left(\frac{S}{c_v \rho}\right), & e(\rho, S) &= c_v \rho^{\gamma-1} \exp\left(\frac{S}{c_v \rho}\right), \\ E(\rho, \mathbf{m}, S) &= \begin{cases} \frac{|\mathbf{m}|^2}{2\rho} + c_v \rho^\gamma \exp\left(\frac{S}{c_v \rho}\right), & \rho > 0, \\ 0, & \rho = 0, \mathbf{m} = \mathbf{0}, S \leq 0, \\ +\infty, & \text{otherwise.} \end{cases} \end{aligned}$$

In particular, $E(\rho, \mathbf{m}, S)$ is understood as the lower-semicontinuous extension of the total energy density to the vacuum states.

DEFINITION 2.1 (Consistent approximation). A sequence $\{\rho_\ell, \mathbf{m}_\ell, S_\ell\}_{\ell=1}^\infty$ is called a consistent approximation of (1.1)–(1.4) in $\Omega \times (0, T)$ if the following conditions are satisfied.

- There exists a sequence of approximate initial data $\{\rho_{0,\ell}, \mathbf{m}_{0,\ell}, S_{0,\ell}\}_{\ell=1}^\infty$ that converges weakly to the initial data in (1.4), namely,

$$\rho_{0,\ell} \rightarrow \rho_0 \text{ weakly in } L^1(\Omega), \quad \mathbf{m}_{0,\ell} \rightarrow \mathbf{m}_0 \text{ weakly in } L^1(\Omega; \mathbb{R}^2), \quad S_{0,\ell} \rightarrow S_0 \text{ weakly in } L^1(\Omega),$$

$$\int_{\Omega} \left(\frac{|\mathbf{m}_{0,\ell}|^2}{2\rho_{0,\ell}} + \rho_{0,\ell} e(\rho_{0,\ell}, S_{0,\ell}) \right) dx dy \rightarrow \int_{\Omega} \left(\frac{|\mathbf{m}_0|^2}{2\rho_0} + \rho_0 e(\rho_0, S_0) \right) dx dy.$$

Moreover, the **energy inequality**

$$\int_{\Omega} \left(\frac{|\mathbf{m}_\ell|^2}{2\rho_\ell} + \rho_\ell e(\rho_\ell, S_\ell) \right) (\cdot, t) dx dy \leq \int_{\Omega} \left(\frac{|\mathbf{m}_{0,\ell}|^2}{2\rho_{0,\ell}} + \rho_{0,\ell} e(\rho_{0,\ell}, S_{0,\ell}) \right) dx dy + e_\ell^1$$

holds for a.e. $t \in [0, T]$, where $e_\ell^1 \rightarrow 0$ as $\ell \rightarrow \infty$;

- The **equation of continuity**

$$\int_0^T \int_{\Omega} (\rho_\ell \varphi_t + \mathbf{m}_\ell \cdot \nabla \varphi) dx dy dt = - \int_{\Omega} \rho_{0,\ell} \varphi(\cdot, 0) dx dy + e_\ell^2[\varphi]$$

holds for every $\varphi \in C_c^1(\Omega \times [0, T])$, with $e_\ell^2[\varphi] \rightarrow 0$ as $\ell \rightarrow \infty$ for every $\varphi \in C_c^2(\Omega \times [0, T])$;

- The **momentum equation**

$$\int_0^T \int_{\Omega} \left(\mathbf{m}_\ell \cdot \varphi_t + \mathbb{1}_{\rho_\ell > 0} \frac{\mathbf{m}_\ell \otimes \mathbf{m}_\ell}{\rho_\ell} : \nabla \varphi + \mathbb{1}_{\rho_\ell > 0} p(\rho_\ell, S_\ell) \nabla \cdot \varphi \right) dx dy dt = - \int_{\Omega} \mathbf{m}_{0,\ell} \cdot \varphi(\cdot, 0) dx dy + e_\ell^3[\varphi]$$

holds for every $\varphi \in C_c^1(\Omega \times [0, T]; \mathbb{R}^2)$, with $e_\ell^3[\varphi] \rightarrow 0$ as $\ell \rightarrow \infty$ for every $\varphi \in C_c^2(\Omega \times [0, T]; \mathbb{R}^2)$;

- The **entropy inequality**

$$\int_0^T \int_{\Omega} \left(S_\ell \varphi_t + \mathbb{1}_{\rho_\ell > 0} \left(S_\ell \frac{\mathbf{m}_\ell}{\rho_\ell} \right) \cdot \nabla \varphi \right) dx dy dt \leq - \int_{\Omega} S_{0,\ell} \varphi(\cdot, 0) dx dy + e_\ell^4[\varphi]$$

holds for every $\varphi \in C_c^1(\Omega \times [0, T])$, $\varphi \geq 0$, with $e_\ell^4[\varphi] \rightarrow 0$ as $\ell \rightarrow \infty$ for every $\varphi \in C_c^2(\Omega \times [0, T])$;

- The **minimum entropy principle** is satisfied; namely, there exists a constant $\underline{s} \in \mathbb{R}$ such that $S_\ell \geq \rho_\ell \underline{s}$ a.e. in $\Omega \times (0, T)$.

Consistent approximations can be constructed by various approaches, including vanishing-viscosity methods and structure-preserving numerical schemes. In [16], different numerical methods defined on a sequence of refined meshes $\{\mathcal{I}_{h_n}\}_{n=1}^\infty$ were considered, where $h_n > 0$ denotes a mesh parameter. The resulting consistent approximations can be written as

$$\{\rho_\ell, \mathbf{m}_\ell, S_\ell\}_{\ell=1}^\infty = \{\rho_{h_n}, \mathbf{m}_{h_n}, S_{h_n}\}_{n=1}^\infty, \quad h_n \rightarrow 0, \quad \ell = \ell(h_n) \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

The following result was established in [13].

THEOREM 2.2. Let $\{\rho_\ell, \mathbf{m}_\ell, S_\ell\}_{\ell=1}^\infty$ be a family of consistent approximations of (1.1)–(1.4) in the sense of Definition 2.1, such that

$$\begin{aligned} \rho_\ell &\rightarrow \rho && \text{weakly-} (*) \text{ in } L^\infty(0, T; L^\gamma(\Omega)), \\ \mathbf{m}_\ell &\rightarrow \mathbf{m} && \text{weakly-} (*) \text{ in } L^\infty\left(0, T; L^{\frac{2\gamma}{\gamma+1}}(\Omega; \mathbb{R}^2)\right), \\ S_\ell &\rightarrow S && \text{weakly-} (*) \text{ in } L^\infty(0, T; L^\gamma(\Omega)) \end{aligned}$$

as $\ell \rightarrow \infty$. Suppose that the limit $(\rho, \mathbf{m}, S)$ is a weak solution of (1.1)–(1.4). Then, there exists a subsequence $\{\rho_{\ell_k}, \mathbf{m}_{\ell_k}, S_{\ell_k}\}$ such that

$$\rho_{\ell_k} \rightarrow \rho \quad \text{in } L^q(0, T; L^\gamma(\Omega)), \quad \mathbf{m}_{\ell_k} \rightarrow \mathbf{m} \quad \text{in } L^q\left(0, T; L^{\frac{2\gamma}{\gamma+1}}(\Omega; \mathbb{R}^2)\right), \quad S_{\ell_k} \rightarrow S \quad \text{in } L^q(0, T; L^\gamma(\Omega))$$

as $k \rightarrow \infty$ for every $1 \leq q < \infty$.

We next recall the definition of DW solutions and state two convergence results established in [13, 15, 16].

DEFINITION 2.3 (DW solution). *Let the initial data satisfy*

$$\rho_0 \in L^\gamma(\Omega), \quad \mathbf{m}_0 \in L^{\frac{2\gamma}{\gamma+1}}(\Omega; \mathbb{R}^2), \quad S_0 \in L^\gamma(\Omega), \quad \int_{\Omega} E(\rho_0, \mathbf{m}_0, S_0) \, dx \, dy < \infty.$$

We say that $(\rho, \mathbf{m}, S)$ is a DW solution of (1.1)–(1.4) in $\Omega \times [0, T)$, $0 < T < \infty$, if the following properties are satisfied.

- **Regularity**

$$\begin{aligned} \rho &\in C_{\text{weak}}([0, T]; L^\gamma(\Omega)), \quad \mathbf{m} \in C_{\text{weak}}\left([0, T]; L^{\frac{2\gamma}{\gamma+1}}(\Omega; \mathbb{R}^2)\right), \\ S &\in L^\infty(0, T; L^\gamma(\Omega)) \cap BV_{\text{weak}}([0, T]; L^\gamma(\Omega)), \\ \int_{\Omega} E(\rho, \mathbf{m}, S)(\cdot, t) \, dx \, dy &\leq \int_{\Omega} E(\rho_0, \mathbf{m}_0, S_0) \, dx \, dy \quad \text{for every } t \in [0, T]. \end{aligned}$$

- **The equation of continuity**

$$\int_0^T \int_{\Omega} (\rho \varphi_t + \mathbf{m} \cdot \nabla \varphi) \, dx \, dy \, dt = - \int_{\Omega} \rho_0 \varphi(\cdot, 0) \, dx \, dy$$

holds for every $\varphi \in C_c^1(\Omega \times [0, T))$.

- **The momentum equation**

$$\int_0^T \int_{\Omega} \left(\mathbf{m} \cdot \varphi_t + \mathbb{1}_{\rho>0} \frac{\mathbf{m} \otimes \mathbf{m}}{\rho} : \nabla \varphi + p(\rho, S) \nabla \cdot \varphi \right) \, dx \, dy \, dt = - \int_0^T \int_{\Omega} \nabla \varphi : d\mathfrak{R}(t) \, dt - \int_{\Omega} \mathbf{m}_0 \cdot \varphi(\cdot, 0) \, dx \, dy$$

holds for every $\varphi \in C_c^1(\Omega \times [0, T); \mathbb{R}^2)$, where $\mathfrak{R} \in L^\infty(0, T; \mathcal{M}^+(\Omega; \mathbb{R}_{\text{sym}}^{2 \times 2}))$ is the Reynolds defect stress.

- **The entropy inequality**

$$\int_{\Omega} \left[S(\cdot, \tau_2+) \varphi(\cdot, \tau_2+) - S(\cdot, \tau_1-) \varphi(\cdot, \tau_1-) \right] \, dx \, dy \geq \int_{\tau_1}^{\tau_2} \int_{\Omega} \left(S \varphi_t + \left\langle \mathcal{V}_{x,y,t}; \mathbb{1}_{\hat{\rho}>0} \left(\hat{S} \frac{\widehat{\mathbf{m}}}{\hat{\rho}} \right) \right\rangle \cdot \nabla \varphi \right) \, dx \, dy \, dt$$

holds, with $S(\cdot, 0-) = S_0$, for every $0 \leq \tau_1 \leq \tau_2 < T$ and every $\varphi \in C_c^1(\Omega \times [0, T))$, $\varphi \geq 0$. Here, $\mathcal{V} \equiv \{\mathcal{V}_{x,y,t}\}_{(x,y,t) \in \Omega \times (0,T)}$ is a parameterized probability (Young) measure satisfying

$$\begin{aligned} \mathcal{V} &\in L^\infty(\Omega \times (0, T); \mathcal{P}(\mathbb{R}^4)), \\ \mathbb{R}^4 &= \left\{ (\hat{\rho}, \widehat{\mathbf{m}}, \widehat{S}) : \hat{\rho} \in \mathbb{R}, \widehat{\mathbf{m}} \in \mathbb{R}^2, \widehat{S} \in \mathbb{R} \right\}, \\ \langle \mathcal{V}_{x,y,t}; \hat{\rho} \rangle &= \rho, \quad \langle \mathcal{V}_{x,y,t}; \widehat{\mathbf{m}} \rangle = \mathbf{m}, \quad \langle \mathcal{V}_{x,y,t}; \widehat{S} \rangle = S. \end{aligned}$$

- **The energy inequality and compatibility between the energy and Reynolds stress defects**

$$\int_{\Omega} E(\rho(\cdot, t), \mathbf{m}(\cdot, t), S(\cdot, t)) \, dx \, dy + \int_{\Omega} d\mathfrak{E}(t) \leq \int_{\Omega} E(\rho_0, \mathbf{m}_0, S_0) \, dx \, dy,$$

where $\mathfrak{E} \in L^\infty(0, T; \mathcal{M}^+(\Omega))$ is the energy defect satisfying

$$2 \min\{1, \gamma - 1\} \mathfrak{E} \leq \text{trace}[\mathfrak{R}] \leq 2 \max\{1, \gamma - 1\} \mathfrak{E}.$$

Remark 2.4. A DW solution $(\rho, \mathbf{m}, S)$ can be interpreted as the first moment of a Young measure $\{\mathcal{V}_{x,y,t}\}$ generated by a consistent approximation of (1.1)–(1.4); see [16, Chapter 5]. Moreover, the dissipative–strong uniqueness principle holds for DW solutions. Consequently, as long as a strong solution exists, every DW solution coincides with it.

THEOREM 2.5 ($\mathcal{K}$ -convergence). *Let $\{\rho_{0,\ell}, \mathbf{m}_{0,\ell}, E_{0,\ell}\}_{\ell=1}^\infty$ satisfy*

$$\rho_{0,\ell} > 0, \quad E_{0,\ell} - \frac{|\mathbf{m}_{0,\ell}|^2}{2\rho_{0,\ell}} > 0, \quad \forall \ell.$$

Let $\{\rho_\ell, \mathbf{m}_\ell, S_\ell\}_{\ell=1}^\infty$ be a consistent approximation of (1.1)–(1.4) in the sense of Definition 2.1. Assume that it is uniformly bounded; namely, there exists a positive constant C such that

$$\|(\rho_\ell, \mathbf{m}_\ell, S_\ell)\|_{L^\infty(\Omega \times (0, T))} \leq C, \quad \forall \ell.$$

Then, there exists a subsequence $\{\rho_{\ell_k}, \mathbf{m}_{\ell_k}, S_{\ell_k}\}$ whose Cesàro averages converge strongly to a DW solution $(\rho, \mathbf{m}, S)$ in the following sense:

$$\begin{aligned} (\tilde{\rho}_n, \tilde{\mathbf{m}}_n, \tilde{S}_n) &:= \frac{1}{n} \sum_{k=1}^n (\rho_{\ell_k}, \mathbf{m}_{\ell_k}, S_{\ell_k}) \rightarrow (\rho, \mathbf{m}, S) \quad \text{in } L^q(\Omega \times (0, T); \mathbb{R}^4), \\ \tilde{E}_n &:= \frac{1}{n} \sum_{k=1}^n E(\rho_{\ell_k}, \mathbf{m}_{\ell_k}, S_{\ell_k}) \rightarrow \left\langle \mathcal{V}_{x,y,t}; E(\hat{\rho}, \hat{\mathbf{m}}, \hat{S}) \right\rangle \quad \text{in } L^q(\Omega \times (0, T)) \end{aligned}$$

as $n \rightarrow \infty$ for every $q \in [1, \infty)$.

3. Numerical Schemes. In this section, we present a unified framework for the first-, second-, third-, fifth-, seventh-, and ninth-order numerical schemes used in the numerical experiments. In particular, we describe the Young-measure schemes and the two families of classical LLF flux-splitting schemes used for comparison.

We first rewrite the 2-D Euler equations of gas dynamics (1.1) in the vector form

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x + \mathbf{G}(\mathbf{U})_y = \mathbf{0},$$

where $\mathbf{U} = (\rho, \rho u, \rho v, E)^\top$, $\mathbf{F}(\mathbf{U}) = (\rho u, \rho u^2 + p, \rho uv, u(E + p))^\top$, and $\mathbf{G}(\mathbf{U}) = (\rho v, \rho uv, \rho v^2 + p, v(E + p))^\top$. Assume that (x_j, y_k) are the points of a uniform Cartesian grid satisfying $x_{j+1} - x_j \equiv \Delta x$ and $y_{k+1} - y_k \equiv \Delta y$. We denote the corresponding numerical solution by $\mathbf{U}_{j,k}(t) \approx \mathbf{U}(x_j, y_k, t)$. The point values $\mathbf{U}_{j,k}$ are evolved in time by numerically solving the following system of ODEs:

$$(3.1) \quad \frac{d\mathbf{U}_{j,k}}{dt} = - \frac{\mathcal{F}_{j+\frac{1}{2},k}^{(r)} - \mathcal{F}_{j-\frac{1}{2},k}^{(r)}}{\Delta x} - \frac{\mathcal{G}_{j,k+\frac{1}{2}}^{(r)} - \mathcal{G}_{j,k-\frac{1}{2}}^{(r)}}{\Delta y},$$

where $r \in \{1, 2, 3, 5, 7, 9\}$ is the prescribed order of spatial accuracy. Note that all of the indexed quantities are time-dependent, but from here on, we will suppress this dependence for the sake of brevity.

The numerical fluxes $\mathcal{F}_{j+\frac{1}{2},k}^{(r)}$ and $\mathcal{G}_{j,k+\frac{1}{2}}^{(r)}$ are constructed dimension by dimension [22]. More precisely, the flux values are first split using the LLF flux splitting and are then reconstructed at the corresponding cell interfaces. Let $\mathcal{R}^{(r),+}$ and $\mathcal{R}^{(r),-}$ denote the left- and right-biased reconstruction operators of order r , respectively. The numerical fluxes are defined by

$$(3.2) \quad \begin{aligned} \mathcal{F}_{j+\frac{1}{2},k}^{(r)} &= \mathcal{R}_{j+\frac{1}{2},k}^{(r),+} \left(\left\{ \mathbf{F}_{\ell,k}^+ \right\}_{\ell \in \mathcal{S}_{j+\frac{1}{2}}^{(r)}} \right) + \mathcal{R}_{j+\frac{1}{2},k}^{(r),-} \left(\left\{ \mathbf{F}_{\ell,k}^- \right\}_{\ell \in \mathcal{S}_{j+\frac{1}{2}}^{(r)}} \right), \\ \mathcal{G}_{j,k+\frac{1}{2}}^{(r)} &= \mathcal{R}_{j,k+\frac{1}{2}}^{(r),+} \left(\left\{ \mathbf{G}_{j,\ell}^+ \right\}_{\ell \in \mathcal{S}_{k+\frac{1}{2}}^{(r)}} \right) + \mathcal{R}_{j,k+\frac{1}{2}}^{(r),-} \left(\left\{ \mathbf{G}_{j,\ell}^- \right\}_{\ell \in \mathcal{S}_{k+\frac{1}{2}}^{(r)}} \right). \end{aligned}$$

Here, the split fluxes are

$$(3.3) \quad \mathbf{F}_{\ell,k}^{\pm} = \frac{1}{2} \left(\Phi_{\ell,k} \pm a_{j+\frac{1}{2},k}^x \mathbf{U}_{\ell,k} \right), \quad \ell \in \mathcal{S}_{j+\frac{1}{2}}^{(r)} \quad \text{and} \quad \mathbf{G}_{j,\ell}^{\pm} = \frac{1}{2} \left(\Psi_{j,\ell} \pm a_{j,k+\frac{1}{2}}^y \mathbf{U}_{j,\ell} \right), \quad \ell \in \mathcal{S}_{k+\frac{1}{2}}^{(r)},$$

where $\Phi_{j,k}$ and $\Psi_{j,k}$ are the unsplit flux values, which will be specified below. The local speeds in (3.3) are estimated by

$$a_{j+\frac{1}{2},k}^x = \max_{\ell \in \mathcal{S}_{j+\frac{1}{2}}^{(r)}} (|u_{\ell,k}| + c_{\ell,k}), \quad a_{j,k+\frac{1}{2}}^y = \max_{\ell \in \mathcal{S}_{k+\frac{1}{2}}^{(r)}} (|v_{j,\ell}| + c_{j,\ell}),$$

where the speed of sound is determined by $c_{j,k} = \sqrt{\frac{\gamma p_{j,k}}{\rho_{j,k}}}$. The corresponding reconstruction stencils are

$$\mathcal{S}_{j+\frac{1}{2}}^{(r)} := \begin{cases} \{j, j+1\}, & r = 1, \\ \{j-1, j, j+1, j+2\}, & r = 2, \\ \{j-s+1, \dots, j+s\}, & r = 2s-1, \quad s = 2, 3, 4, 5, \end{cases}$$

and $\mathcal{S}_{k+\frac{1}{2}}^{(r)}$ is defined analogously in the y -direction.

For systems of conservation laws, the reconstructions of order $r \geq 2$ are performed in local characteristic variables, a strategy that is often used in high-order schemes to reduce spurious oscillations and can also enhance the resolution of second-order schemes; see, e.g., [23, 25, 28–30] and references therein. In particular, each operator in (3.2) includes the projection of all split fluxes on the stencil onto a single characteristic basis constructed at the corresponding interface, the componentwise scalar reconstruction in the characteristic variables, and the transformation of the reconstructed fluxes back to the conservative variables. The detailed formulas are provided in Appendix A.

3.1. Young-Measure Approach. In the schemes using Young measures, a discrete measure is reconstructed at every spatial grid point and at every evaluation of the right-hand side of (3.1). Let $\mathbf{z} = (z^{(1)}, \dots, z^{(d)})^\top \in \mathbb{R}^d$, with $d = 4$ in the 2-D Euler equations of gas dynamics, denote a generic phase-space variable, namely, a possible value of the conservative vector $\mathbf{U}$. The phase-space grid points are denoted by $\mathbf{z}_\ell \in \mathbb{R}^d$, $\ell = 1, \dots, N_{\mathbf{u}}$. Here, $N_{\mathbf{u}}$ denotes the total number of phase-space grid points. If N_q grid points are used in the q -th phase-space direction before the admissibility restriction described below, then the underlying tensor-product grid contains $\prod_{q=1}^d N_q$ points. If a uniform tensor-product grid is used in phase space, its cell volume is $\Delta V = \prod_{q=1}^d \Delta z^{(q)}$. We assume that the phase-space discretization is chosen so that the constraints introduced below are feasible; in particular, every evolved state $\mathbf{U}_{j,k}$ must belong to the convex hull of the phase-space grid points.

At each grid point (x_j, y_k) , the values $\{\mu_{j,k,\ell}^*\}_{\ell=1}^{N_{\mathbf{u}}}$ are obtained by solving the LP problem

$$(3.4) \quad \{\mu_{j,k,\ell}^*\}_{\ell=1}^{N_{\mathbf{u}}} \in \underset{\{\mu_\ell\}_{\ell=1}^{N_{\mathbf{u}}}}{\operatorname{argmin}} \Delta V \sum_{\ell=1}^{N_{\mathbf{u}}} \eta(\mathbf{z}_\ell) \mu_\ell,$$

$$(3.5) \quad \text{subject to} \quad \mu_\ell \geq 0, \quad \ell = 1, \dots, N_{\mathbf{u}},$$

$$(3.6) \quad \mu_\ell \leq \frac{1}{\Delta V}, \quad \ell = 1, \dots, N_{\mathbf{u}},$$

$$(3.7) \quad \Delta V \sum_{\ell=1}^{N_{\mathbf{u}}} \mu_\ell = 1,$$

$$(3.8) \quad \Delta V \sum_{\ell=1}^{N_{\mathbf{u}}} \mathbf{z}_\ell \mu_\ell = \mathbf{U}_{j,k}.$$

Thus, $\nu_{j,k}^* = \Delta V \sum_{\ell=1}^{N_{\mathbf{u}}} \mu_{j,k,\ell}^* \delta_{\mathbf{z}_\ell}$ is a discrete probability measure whose first moment $\Delta V \sum_{\ell=1}^{N_{\mathbf{u}}} \mathbf{z}_\ell \mu_{j,k,\ell}^*$ coincides with the evolved conservative state $\mathbf{U}_{j,k}$. Notice that the upper bound in (3.6) follows from the nonnegativity and normalization constraints (3.5) and (3.7); it is retained here to match the LP problem used in the implementation.

In this paper, the objective function in (3.4) is the mathematical entropy, and consequently, for the prescribed first moment $\mathbf{U}_{j,k}$, minimizing the objective function in (3.4) is equivalent to maximizing the expected physical entropy over all admissible discrete measures satisfying the constraints (3.5)–(3.8). After the minimizer $\mu_{j,k,\ell}^*$ has been determined, the flux values in (3.3) are computed by averaging the physical fluxes with respect to $\mu_{j,k,\ell}^*$:

$$\Phi_{j,k}^{\text{YM}} := \langle \nu_{j,k}^*, \mathbf{F} \rangle = \Delta V \sum_{\ell=1}^{N_u} \mathbf{F}(z_\ell) \mu_{j,k,\ell}^*, \quad \Psi_{j,k}^{\text{YM}} := \langle \nu_{j,k}^*, \mathbf{G} \rangle = \Delta V \sum_{\ell=1}^{N_u} \mathbf{G}(z_\ell) \mu_{j,k,\ell}^*.$$

For the Young-measure schemes, we set $\Phi_{j,k} = \Phi_{j,k}^{\text{YM}}$ and $\Psi_{j,k} = \Psi_{j,k}^{\text{YM}}$ in (3.3). For $r \geq 2$, the characteristic basis used in the Young-measure schemes is constructed from the arithmetic averages of the primitive variables, as described in Appendix A. The left and right eigenvector matrices used for the LCD are therefore not averaged with respect to the Young measure.

3.2. Classical LLF Flux-Splitting Schemes. For the classical LLF flux-splitting schemes, the unsplit flux values in (3.3) are evaluated directly by the evolved conservative state:

$$\Phi_{j,k} = \mathbf{F}(\mathbf{U}_{j,k}), \quad \Psi_{j,k} = \mathbf{G}(\mathbf{U}_{j,k}).$$

All other components of the discretization remain unchanged. We consider two choices of the characteristic basis. In the standard LLF schemes, the basis is constructed using the arithmetic averages of the primitive variables, as described in Appendix A. In the LLF-ELCD schemes, the interface state used to construct the characteristic basis is selected from four candidate states according to their physical entropy; see Appendix B.

It is important to distinguish the entropy-based LCD from the LP closure in (3.4)–(3.8). The former uses entropy only to select an interface state for the LCD and does not optimize the numerical solution over a class of admissible measures. It therefore does not enforce a maximum-entropy selection principle. Moreover, since no LCD is needed when $r = 1$, the two classical first-order LLF schemes coincide.

Remark 3.1. The semi-discrete ODE system (3.1) is integrated in time using the three-stage third-order strong stability preserving Runge-Kutta method (see, e.g., [20, 21]) for the second-, third-, fifth-, seventh-, and ninth-order schemes and the forward Euler method for the first-order scheme. For the Young-measure schemes, the LP closure is recomputed at every Runge-Kutta stage. In all of the numerical experiments, we take the CFL number to be 0.475 and $\gamma = 1.4$. In the rest of the paper, we refer to the schemes of formal order $r \in \{1, 2, 3, 5, 7, 9\}$ in the three families—the standard LLF flux-splitting schemes, the LLF flux-splitting schemes equipped with the entropy-based LCD, and the Young-measure schemes—as the r -Order LLF, r -Order LLF-ELCD, and r -Order YM schemes, respectively, for the sake of brevity.

Remark 3.2. The LP problem (3.4)–(3.8) can be solved using the black-box solver `linprog` in MATLAB or the KKT-based solver described, for example, in [5, Appendix D].

4. Numerical Results. In this section, we apply the schemes described in §3 to the KH instability problem. All computations are performed on the domain $[0, 1] \times [0, 1]$ using a uniform mesh with $\Delta x = \Delta y = 1/512$.

To facilitate comparisons across formal order of the schemes, we use the following notation for the solutions computed within each scheme family, but with different orders of the resulting scheme: $(\rho_\ell, \mathbf{m}_\ell, S_\ell)$, $\ell = 1, \dots, 6$, are solutions computed by the first-, second-, third-, fifth-, seventh-, and ninth-order schemes, respectively, that is, $\ell(1) = 1$, $\ell(2) = 2$, $\ell(3) = 3$, $\ell(5) = 4$, $\ell(7) = 5$, and $\ell(9) = 6$. Motivated by the concept of $\mathcal{K}$ -convergence described in Theorem 2.5, we will average these solutions to obtain $(\tilde{\rho}_n, \tilde{\mathbf{m}}_n, \tilde{S}_n)^\top := \frac{1}{n} \sum_{\ell=1}^n (\rho_\ell, \mathbf{m}_\ell, S_\ell)^\top$, where $n = 1, \dots, 6$.

As explained in Remark 2.4, a sequence of approximate solutions that converges strongly to a weak solution generates a Young measure concentrated at a single state, that is, a Dirac mass. By contrast, a non-Dirac Young measure can record oscillations that persist under weak convergence. To examine this distinction numerically, we consider a local empirical distribution of the computed density. At a fixed time, this distribution is constructed from the density values $\rho_{j,k}$ at all grid points in the small rectangular subdomain $\hat{\Omega}$ introduced below. It therefore measures the spatial variation of the numerical density

within this region and provides a common regional diagnostic for the three scheme families. This spatial histogram is different from the pointwise density marginal, which describes the distribution of density states at a fixed space-time point. In particular, the spread of the histogram may reflect both small-scale oscillations and macroscopic spatial variation within $\tilde{\Omega}$. Its direct identification with the pointwise density marginal would require additional locality assumptions, such as sufficient scale separation or an appropriate shrinking-neighborhood limit. We therefore use the spatial histogram only to compare the regional density distributions produced by the different schemes, rather than as evidence that the limiting Young measure is non-Dirac.

To this end, we take a certain small rectangular subdomain $\tilde{\Omega}$ and compute

$$\underline{\rho} := \min_{(x_j, y_k) \in \tilde{\Omega}} \rho_{j,k} \quad \text{and} \quad \bar{\rho} := \max_{(x_j, y_k) \in \tilde{\Omega}} \rho_{j,k}.$$

For each comparison, the same values of $\underline{\rho}$, $\bar{\rho}$, and the same bin edges are used for all distributions being compared. We then introduce $\Delta\rho := (\bar{\rho} - \underline{\rho})/30$ and the following partition of the interval $[\underline{\rho}, \bar{\rho}]$:

$$[\underline{\rho}, \bar{\rho}] = \bigcup_{i=1}^{30} [\rho^{(i-\frac{1}{2})}, \rho^{(i+\frac{1}{2})}), \quad \text{where} \quad \rho^{(\frac{1}{2})} = \underline{\rho}, \quad \rho^{(i+\frac{1}{2})} = \rho^{(i-\frac{1}{2})} + \Delta\rho, \quad i = 1, \dots, 30.$$

The distribution of the dataset formed by the values $\rho_{j,k}$ at all grid points $(x_j, y_k) \in \tilde{\Omega}$ is represented by a function $\sigma_\ell = \sigma_\ell(\rho)$, where the subscript $\ell = \ell(r)$ indicates that the density values have been computed using the r -th order scheme. More specifically, $\sigma_\ell(\rho^{(i)})$ denotes the number of grid points (x_j, y_k) for which the computed values $\rho_{j,k}$ fall within the interval $[\rho^{(i-\frac{1}{2})}, \rho^{(i+\frac{1}{2})})$. Additionally, we apply a normalization such that

$$\Delta\rho \sum_{i=1}^{30} \sigma_\ell(\rho^{(i)}) = 1.$$

Motivated again by $\mathcal{K}$ -convergence, we average the sequence $\{\sigma_\ell\}$ and introduce $\tilde{\sigma}_n := \frac{1}{n} \sum_{\ell=1}^n \sigma_\ell$, $n = 1, \dots, 6$.

We consider the KH instability problem taken from [13] with the following initial conditions:

$$(\rho, u, v, p)(x, y, 0) = \begin{cases} (2, -0.5, 0, 2.5), & \text{if } 0.25 + 0.05Y_1(x, \omega) < y < 0.75 + 0.05Y_2(x, \omega), \\ (1, 0.5, 0, 2.5), & \text{otherwise,} \end{cases}$$

where

$$Y_j(x, \omega) = \sum_{n=1}^{10} a_j^n(\omega) \cos(b_j^n(\omega) + 2n\pi x), \quad j = 1, 2,$$

with $a_j^n = a_j^n(\omega) \in [0, 1]$ and $b_j^n = b_j^n(\omega) \in (-\pi, \pi)$ being fixed random numbers, and with a_j^n normalized so that $\sum_{n=1}^{10} a_j^n = 1$. The particular values of a_j^n and b_j^n were generated by `rand` in MATLAB and are reported in [7, Table 1]. The boundary conditions in this example are periodic. In all the Young-measure schemes, we set $[0.2, 3.2] \times [-2, 2] \times [-1.5, 1.5] \times [2, 10]$ for the discretization in phase space and use 20 grid points in each phase-space variable before retaining only physically admissible nodes. Accordingly, $N_{\mathbf{u}}$ denotes the total number of retained phase-space grid points.

4.1. Numerical Solutions and Approximate Young Measures. We compute the numerical solutions using all the studied schemes until the final time $T = 2$ and present in Figure 4.1 the results obtained by the 3-Order, 5-Order, 7-Order and 9-Order schemes. The results show that the shear layers roll up into vortices, with the higher-order schemes resolving an increasing number of small-scale structures. Moreover, the differences between the LLF and LLF-ELCD schemes are quite limited, while the differences between these two families and the YM schemes are noticeable in the resulting vortical patterns, especially for higher-order schemes. Consequently, at this fixed mesh resolution, the numerical approximations do not appear to stabilize toward a common flow pattern as the formal order increases. Although this finite-order comparison does not establish convergence to distinct DW solutions or rule out strong convergence,

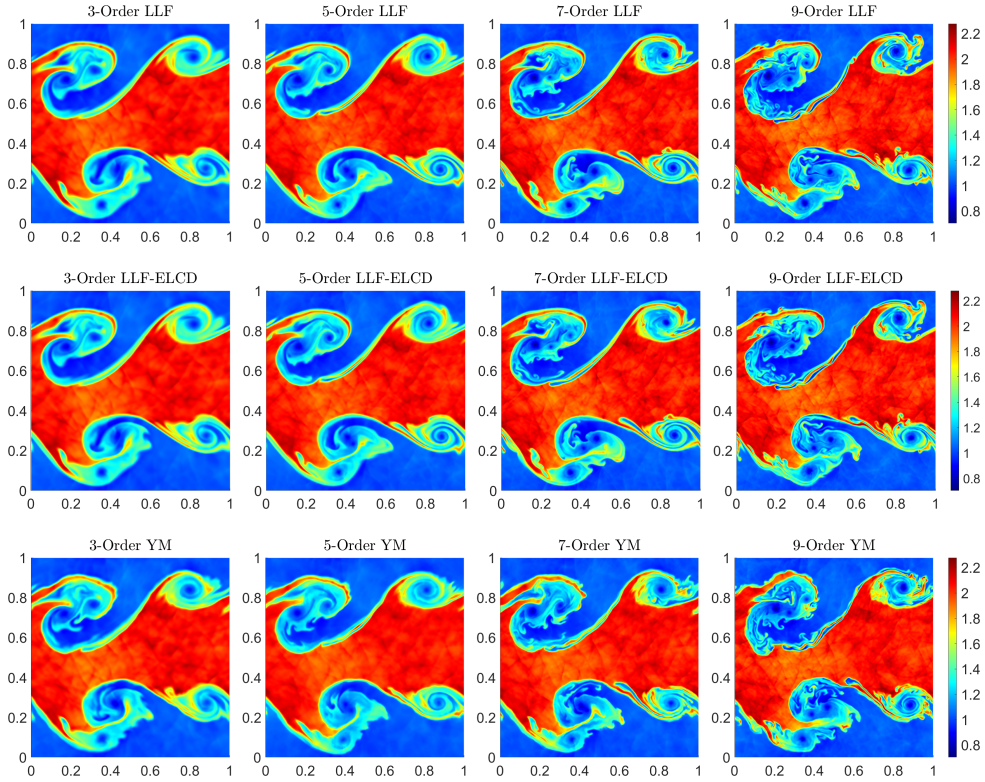


FIG. 4.1. KH Instability: Density computed by the 3-Order (left column), 5-Order (second column), 7-Order (third column), and 9-Order (right column) LLF (top row), LLF-ELCD (middle row), and YM (bottom row) schemes.

it provides evidence of scheme-dependent numerical selection. The cumulative averages and empirical density distributions introduced above will therefore be used to investigate this behavior in greater detail.

We consider the finite cumulative averages of the numerical solutions over different orders. In Figure 4.2, we present $\tilde{\rho}_3$, $\tilde{\rho}_4$, $\tilde{\rho}_5$, and $\tilde{\rho}_6$, where one can see that, in contrast to Figure 4.1, the average density profiles show similarities for different n , indicating increased stability as n increases. Additionally, we observe that compared to the LLF and LLF-ELCD schemes, the average density profiles produced by the YM schemes differ significantly. This suggests scheme-dependent numerical selection, although it does not prove convergence to distinct DW solutions.

Next, we approximate the time averages of a DW solution by evaluating the time averages of the computed densities, which are denoted by $\rho_\ell^T := \frac{1}{T} \int_0^T \rho_\ell(\cdot, t) dt$, where T is the final time. We report ρ_ℓ^T for $\ell = 3, 4, 5, 6$ computed by the three studied schemes in Figure 4.3, where one can observe that they become visually more stable as the order increases. In addition, the differences between the time-averaged flows are substantially smaller than those between the instantaneous solutions. In Figure 4.4, we show the time averages of $\tilde{\rho}_6$, which are denoted by $\tilde{\rho}_6^T := \frac{1}{T} \int_0^T \tilde{\rho}_6(\cdot, t) dt$. The three time-averaged profiles are remarkably similar. This phenomenon suggests consistency in the measure-valued solutions produced by different numerical methods, but does not establish that the three methods generate the same measure-valued limit.

To illustrate the differences in numerical results, we also compare the regional probability density functions (PDFs) computed by the three studied schemes. To this end, we select two subdomains:

$$\tilde{\Omega}_1 = [0.24, 0.26] \times [0.80, 0.82] \quad \text{and} \quad \tilde{\Omega}_2 = [0.36, 0.38] \times [0.24, 0.26],$$

and present the obtained results in Figure 4.5. It is evident that these PDFs computed by the LLF and LLF-ELCD schemes have similarities, especially in the subdomain $\tilde{\Omega}_2$, while those computed by the YM

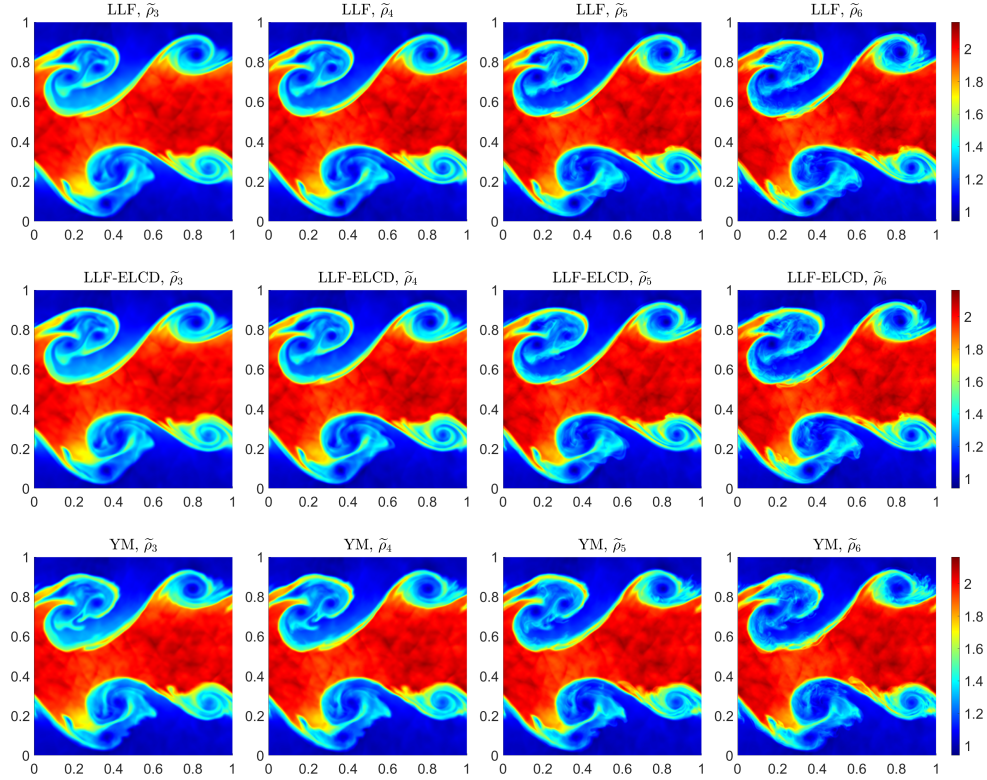


FIG. 4.2. KH Instability: $\tilde{\rho}_3$ (left column), $\tilde{\rho}_4$ (second column), $\tilde{\rho}_5$ (third column), and $\tilde{\rho}_6$ (right column) computed by the LLF (top row), LLF-ELCD (middle row), and YM (bottom row) schemes.

method differ significantly from the other two. This provides additional evidence that different methods may select different DW solutions.

4.2. Comparison Based on Selection Criteria. According to [7, 14, 17], possible selection criteria for DW solutions are (i) the maximization of entropy production, (ii) the minimization of energy defect, (iii) the maximization of energy defect, and (iv) the minimization of the Bregman distance. Let us briefly discuss the physical interpretation of the proposed selection criteria. First, criteria (i) and (iv) are related to the Second Law of Thermodynamics, which states that physically reliable solutions either preserve or increase the entropy over time. Here, we select the solutions that either maximize the entropy production (i) or are closest to the equilibrium state with maximal entropy (iv). The latter expresses the Clausius formulation of the Second Law of Thermodynamics: “The energy of the world (closed system) is constant; its entropy tends to a maximum.” Criteria (ii) and (iii) are not contradictory since they select solutions with either the minimal or maximal oscillatory (turbulent) structures, respectively. As in Definition 2.3, the energy defect controls the turbulent Reynolds stress defect. As shown in [14], the class of DW solutions emanating from given initial data is convex, closed, and weakly compact in a suitable topology. The energy $E(\rho, \mathbf{m}, S)$ is a strictly convex function on its domain of definition with $\rho > 0$. Consequently, the energy defect is a strictly concave function and criteria (iii) and (iv) yield a unique selected solution in a DW class.

We first examine the entropy-related selection criterion. In the LLF-ELCD schemes, the entropy-based LCD uses physical entropy only to select the interface state that defines the local characteristic basis; see Appendix B. In the YM schemes, the expected mathematical entropy is minimized in the LP problem (3.4)–(3.8) to compute the discrete probability measures. We therefore examine whether these two uses of entropy lead to larger physical entropy in the computed solutions. In Figure 4.6, we show the differences in the entropy at the final time among the LLF, LLF-ELCD, and YM schemes. We observe that these differences are generally more pronounced as the formal order increases. For the lower-order schemes, the differences between the LLF and LLF-ELCD results are relatively small. By contrast, the

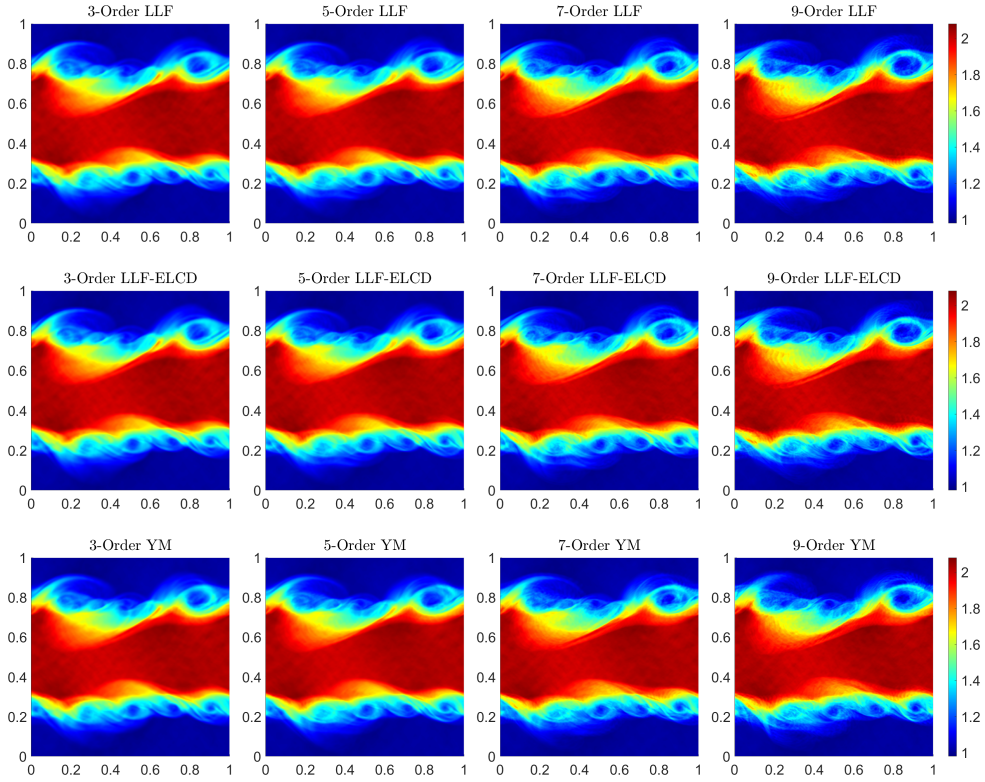


FIG. 4.3. KH Instability: ρ_3^T (left column), ρ_4^T (second column), ρ_5^T (third column), and ρ_6^T (right column) computed by the LLF (top row), LLF-ELCD (middle row), and YM (bottom row) schemes.

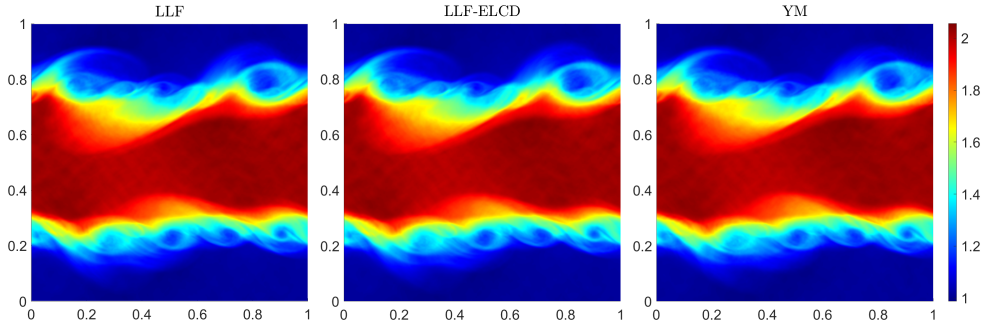


FIG. 4.4. KH Instability: $\tilde{\rho}_6^T$ computed by the LLF (left), LLF-ELCD (middle), and YM (right) schemes.

differences between the LLF and YM results are substantially more pronounced than those between the LLF and LLF-ELCD results.

In Table 4.1, we report the total physical entropy for each formal order r ,

$$\mathcal{S}_r(t) := \int_{\Omega} S_r(x, y, t) \, dx dy,$$

computed by the three schemes studied. Here, the LLF-ELCD schemes do not achieve larger average entropies of different formal orders, while the YM schemes produce larger entropies in **all** of the formal orders. The numerical results demonstrate that using entropy only to select an interface state for the LCD does not optimize the numerical solution and also does not enforce a maximum-entropy selection principle. By contrast, the YM results are consistent with the local maximum-entropy bias imposed through the objective function in the LP problem, but they do not by themselves prove selection of a

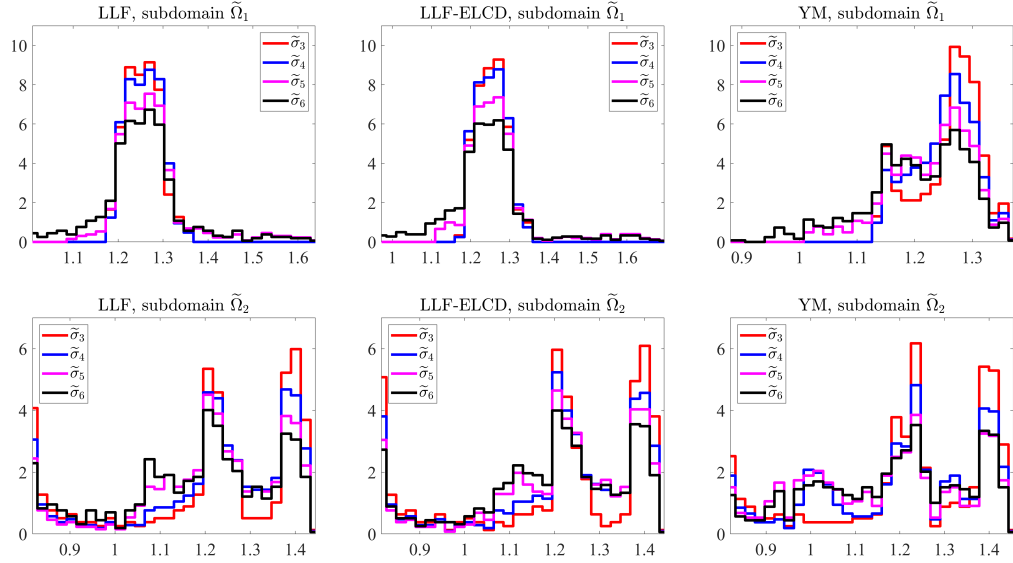


FIG. 4.5. KH Instability: $\tilde{\sigma}_3$, $\tilde{\sigma}_4$, $\tilde{\sigma}_5$, and $\tilde{\sigma}_6$ computed by the LLF (left column), LLF-ELCD (middle column), and YM (right column) schemes in the subdomains $\tilde{\Omega}_1$ (top row) and $\tilde{\Omega}_2$ (bottom row).

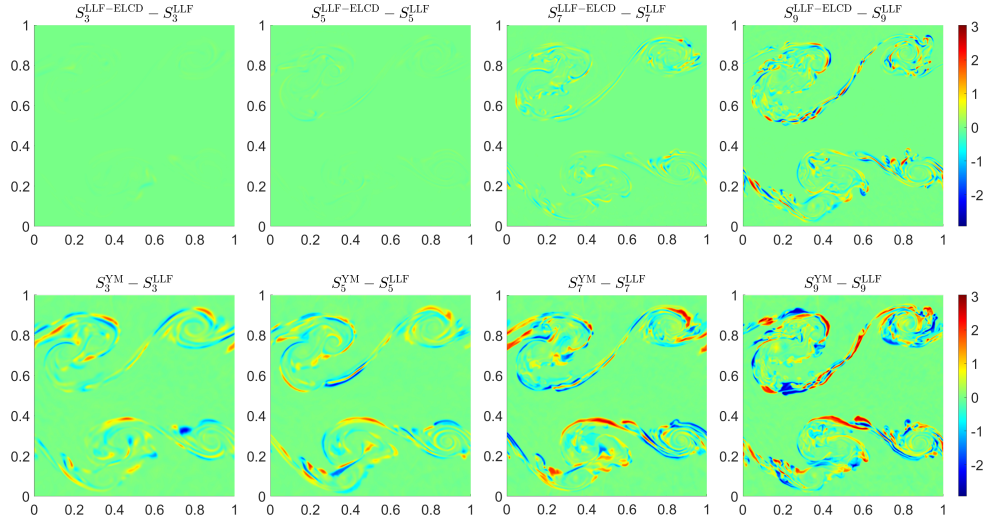


FIG. 4.6. KH Instability: The entropy differences between the 3-Order (left column), 5-Order (second column), 7-Order (third column), and 9-Order (right column) LLF with the corresponding LLF-ELCD (top row) and YM (bottom row) schemes at the final time $T = 2$.

globally maximal-entropy DW solution.

We then compute the average total entropy over orders up to n by

$$\mathcal{S}^n(t) := \int_{\Omega} \tilde{S}_n(x, y, t) dx dy,$$

the averaged total energy and the energy of the cumulative mean state by

$$\tilde{E}_n(x, y, t) := \frac{1}{n} \sum_{\ell=1}^n E(\rho_{\ell}(x, y, t), \mathbf{m}_{\ell}(x, y, t), S_{\ell}(x, y, t)),$$

TABLE 4.1

KH Instability: Time averages of the total physical entropy for the 1-Order, 2-Order, 3-Order, 5-Order, 7-Order, and 9-Order LLF, LLF-ELCD, and YM schemes.

	LLF	LLF-ELCD	YM	Relative improvement of YM over LLF (%)
$\frac{1}{T} \int_0^T \mathcal{S}_1(t) dt$	1.1160477	1.1160477	1.1177841	0.15558
$\frac{1}{T} \int_0^T \mathcal{S}_2(t) dt$	1.0627867	1.0627606	1.0632212	0.04088
$\frac{1}{T} \int_0^T \mathcal{S}_3(t) dt$	1.0617010	1.0616969	1.0623086	0.05723
$\frac{1}{T} \int_0^T \mathcal{S}_5(t) dt$	1.0609902	1.0609699	1.0629257	0.18242
$\frac{1}{T} \int_0^T \mathcal{S}_7(t) dt$	1.0556312	1.0555922	1.0582666	0.24965
$\frac{1}{T} \int_0^T \mathcal{S}_9(t) dt$	1.0528719	1.0527596	1.0543940	0.14457

$$\mathcal{E}_1^n(t) := \int_{\Omega} \tilde{E}_n(x, y, t) dx dy = \int_{\Omega} \frac{1}{n} \sum_{\ell=1}^n \left[\frac{|\mathbf{m}_{\ell}|^2}{2\rho_{\ell}} + \rho_{\ell} e(\rho_{\ell}, S_{\ell}) \right] dx dy,$$

$$\mathcal{E}_2^n(t) := \int_{\Omega} E(\tilde{\rho}_n(x, y, t), \tilde{\mathbf{m}}_n(x, y, t), \tilde{S}_n(x, y, t)) dx dy = \int_{\Omega} \frac{|\tilde{\mathbf{m}}_n|^2}{2\tilde{\rho}_n} + \tilde{\rho}_n e(\tilde{\rho}_n, \tilde{S}_n) dx dy,$$

respectively, and the energy defect is defined by

$$\mathcal{D}_{\text{E}}^n(t) := \int_{\Omega} \left| \tilde{E}_n(x, y, t) - E(\tilde{\rho}_n(x, y, t), \tilde{\mathbf{m}}_n(x, y, t), \tilde{S}_n(x, y, t)) \right| dx dy.$$

In addition, we consider the Bregman distance between a solution $(\rho, \mathbf{m}, S)$ at time t and the constant equilibrium state given by $(\bar{\rho}, \mathbf{0}, S(\bar{\rho}, \bar{\theta}))$ that maximizes the entropy. Here,

$$\bar{\rho} = \frac{1}{|\Omega|} \int_{\Omega} \rho_0 dx dy, \quad \bar{\theta} = \frac{1}{c_v \bar{\rho} |\Omega|} \int_{\Omega} E_0 dx dy,$$

and the Bregman distance is given by

$$\mathcal{J}^n(t) := \mathcal{E}_2^n(t) - \bar{\theta} \mathcal{S}^n(t);$$

see [14] for details on its derivation.

Based on the numerical solutions obtained by different methods, we compare their respective time-averaged total entropy, energy defects, and the Bregman distance to the equilibrium state. The time averages of $\mathcal{S}^6$, $\mathcal{E}_1^6$, $\mathcal{E}_2^6$, $\mathcal{D}_{\text{E}}^6$, and $\mathcal{J}^6$ are presented in Table 4.2. Here, the initial data yield $\bar{\theta} = 1.71665$. It can be observed that the YM schemes produce a smaller energy defect than the LLF and LLF-ELCD schemes. At the same time, these two methods result in smaller time-averaged total entropies. According to these numerical diagnostics, the YM schemes are prioritized by criteria (i), (ii), and (iv), while the LLF-ELCD schemes are prioritized by criterion (iii). Note that the differences between the LLF and LLF-ELCD schemes with respect to all selection criteria are very small.

4.3. Density Marginals of the Averaged Young Measures. At each physical grid point (x_j, y_k) , the coefficients $\mu_{j,k,\ell}^*$ describe the local distribution of the Young measure over the discrete phase-space states $\mathbf{z}_{\ell}$. They satisfy

$$\mu_{j,k,\ell}^* \geq 0, \quad \Delta V \sum_{\ell=1}^{N_u} \mu_{j,k,\ell}^* = 1,$$

where ΔV is the constant phase-space cell volume. Therefore, $\mu_{j,k,\ell}^*$ is a weight of the probability density. We define $w_{j,k,\ell} := \Delta V \mu_{j,k,\ell}^*$. The normalized coefficients $w_{j,k,\ell}$ satisfy $w_{j,k,\ell} \geq 0$ and $\sum_{\ell=1}^{N_u} w_{j,k,\ell} = 1$.

TABLE 4.2

KH Instability: Time averages of the total physical entropy, averaged total energy, energy of the cumulative mean state, the energy defect, and the Bregman distance computed by the LLF, LLF-ELCD, and YM schemes.

	LLF	LLF-ELCD	YM	Relative change of YM over LLF (%)
$\frac{1}{T} \int_0^T \mathcal{S}^6(t) dt$	1.0683381	1.0683045	1.0698167	0.13840
$\frac{1}{T} \int_0^T \mathcal{E}_1^6(t) dt$	6.4375014	6.4375014	6.4375014	0
$\frac{1}{T} \int_0^T \mathcal{E}_2^6(t) dt$	6.4127463	6.4127242	6.4136133	0.01352
$\frac{1}{T} \int_0^T \mathcal{D}_E^6(t) dt$	2.47551e-2	2.47772e-2	2.38881e-2	-3.50231
$\frac{1}{T} \int_0^T \mathcal{J}^6(t) dt$	4.5787794	4.5788151	4.5771083	-0.03650

Accordingly, the discrete Young measure at (x_j, y_k) can be written as $\nu_{j,k}^* = \sum_{\ell=1}^{N_u} w_{j,k,\ell} \delta_{\mathbf{z}_\ell}$. Moreover, its first moment satisfies $\sum_{\ell=1}^{N_u} w_{j,k,\ell} \mathbf{z}_\ell = \mathbf{U}_{j,k}$.

The values of $w_{j,k,\ell}$ provide direct information about the local concentration of the Young measure. If a weight is close to one, the measure is strongly concentrated at a single phase-space state. It is close to a Dirac measure. If several weights have comparable nonzero values, the measure is a locally dispersed Young measure. Therefore, the spatial distributions of $w_{j,k,\ell}$, together with quantities such as $\max_\ell w_{j,k,\ell}$ and the effective number of active states, can be used to examine the concentration of the Young measure.

In the following, we study the density marginals averaged first over a selected physical subdomain and then over several YM reconstruction orders. Thus, the two figures do not display the local full-state weights $w_{j,k,\ell}$. For the YM scheme of order $r \in \{1, 2, 3, 5, 7, 9\}$, we write the density marginal at (x_j, y_k) as $\omega_{j,k}^{(r)} = \sum_{m=1}^{N_\rho} w_{j,k,m}^{\rho^{(r)}} \delta_{\rho_m^*}$, where ρ_m^* is the m -th density support point. The mass $w_{j,k,m}^{\rho^{(r)}}$ is obtained by summing the weights of all phase-space states whose density component equals ρ_m^* . In the numerical output, it is recovered from the saved density-marginal coefficient through $w_{j,k,m}^{\rho^{(r)}} = \Delta \rho \mu_{j,k,m}^{\rho^{(r)}}$, where $\mu_{j,k,m}^{\rho^{(r)}}$ is the density-marginal coefficient. These marginal masses satisfy

$$w_{j,k,m}^{\rho^{(r)}} \geq 0, \quad \sum_{m=1}^{N_\rho} w_{j,k,m}^{\rho^{(r)}} = 1, \quad \sum_{m=1}^{N_\rho} \rho_m^* w_{j,k,m}^{\rho^{(r)}} = \rho_{j,k}^{(r)}.$$

We next perform two successive averages: First, we average over physical cells in a selected spatial subdomain. Let $\mathcal{I}$ denote the corresponding set of grid indices corresponding to the physical cells. For the scheme of order r , we define

$$\bar{w}_m^{\rho^{(r)}} := \frac{1}{|\mathcal{I}|} \sum_{(j,k) \in \mathcal{I}} w_{j,k,m}^{\rho^{(r)}}, \quad m = 1, \dots, N_\rho.$$

The same definition is applied separately to each selected subdomain. Since the local density marginals are normalized, their spatial averages satisfy

$$\bar{w}_m^{\rho^{(r)}} \geq 0, \quad \sum_{m=1}^{N_\rho} \bar{w}_m^{\rho^{(r)}} = 1.$$

We then average the spatially averaged masses obtained with different YM reconstruction orders. To this end, we introduce

$$\mathcal{R}_3 = \{1, 2, 3\}, \quad \mathcal{R}_4 = \{1, 2, 3, 5\}, \quad \mathcal{R}_5 = \{1, 2, 3, 5, 7\}, \quad \text{and} \quad \mathcal{R}_6 = \{1, 2, 3, 5, 7, 9\}.$$

For $q = 3, \dots, 6$, we define

$$\tilde{w}_{q,m}^\rho := \frac{1}{q} \sum_{r \in \mathcal{R}_q} \bar{w}_m^{\rho^{(r)}}, \quad \tilde{\omega}_q := \sum_{m=1}^{N_\rho} \tilde{w}_{q,m}^\rho \delta_{\rho_m^*}.$$

Here, q denotes the number of schemes included in the average and not the formal order of a numerical scheme. Note that every $\tilde{\omega}_q$ is a probability measure on $\mathbb{R}$.

The weights $\tilde{w}_{q,m}^\rho$ are shown in Figure 4.7. For each subdomain, the four curves represent $\tilde{\omega}_3$, $\tilde{\omega}_4$, $\tilde{\omega}_5$, and $\tilde{\omega}_6$. The height of each horizontal segment is $\tilde{w}_{q,m}^\rho$, namely, the mass assigned to the density cell centered at ρ_m^* . Most of the probability mass is concentrated on several neighboring density cells, while the masses outside this region are negligible. The differences among the four averages are mainly caused by a distribution of weights among these cells. These differences indicate that the averaged marginals stabilize as higher orders are included.

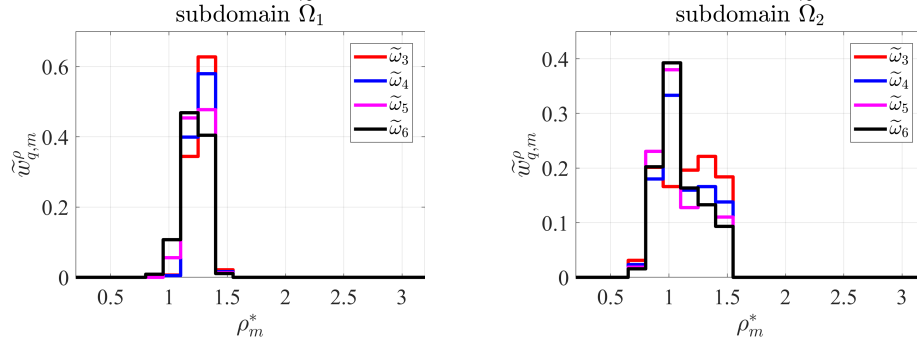


FIG. 4.7. KH Instability: Probability masses $\tilde{w}_{q,m}^\rho$ of the averaged density marginals $\tilde{\omega}_q$, $q = 3, \dots, 6$, in the subdomains $\tilde{\Omega}_1$ (left) and $\tilde{\Omega}_2$ (right).

We have also plotted the corresponding cumulative distribution functions

$$F_q(\rho_m^*) := \sum_{s=1}^m \tilde{w}_{q,s}^\rho, \quad q = 3, \dots, 6,$$

in Figure 4.8. Here, $F_q(\rho_m^*)$ is the probability assigned to density values not exceeding ρ_m^* , and its jump at ρ_m^* is exactly $\tilde{w}_{q,m}^\rho$. All four cumulative distribution functions are nondecreasing and reach one at the largest density support value, confirming the normalization of the averaged density marginals. Their closer agreement for the later averages is consistent with Figure 4.7.

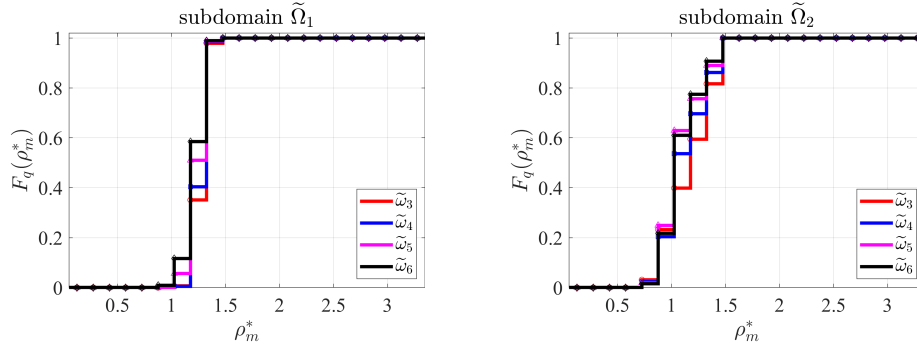


FIG. 4.8. KH Instability: Cumulative distribution functions $F_q(\rho_m^*)$ of the averaged density marginals $\tilde{\omega}_q$, $q = 3, \dots, 6$, in the subdomains $\tilde{\Omega}_1$ (left) and $\tilde{\Omega}_2$ (right).

5. Conclusions. We have extended the linear-programming-based Young-measure schemes to two space dimensions and applied them to the Kelvin–Helmholtz instability governed by the compressible Euler equations. Comparisons with the standard LLF and LLF–ELCD schemes reveal scheme-dependent differences in the small-scale vortical structures and PDFs. We have also examined several possible selection criteria for DW solutions. At every spatial reconstruction order considered, the YM schemes produce the largest time-averaged domain-integrated physical entropy. By contrast, the LLF–ELCD schemes do

not systematically produce larger time-averaged domain-integrated physical entropy than the corresponding standard LLF schemes. Therefore, using entropy only to select an interface state for the LCD does not enforce a maximum-entropy selection principle. The YM results are consistent with the local maximum-entropy bias imposed through the objective function of the linear-programming problem. Moreover, the maximum-entropy, minimum-energy-defect, and minimum-Bregman-functional criteria favor the YM schemes. The finite cross-order and time averages become more stable as additional reconstruction orders are included, while the averaged density marginals remain concentrated on several neighboring density states. These fixed-mesh observations provide evidence of scheme-dependent numerical selection. Overall, the results indicate that the objective function in the linear-programming problem is an active component of the Young-measure closure rather than merely an auxiliary computational device. Future work will investigate alternative objective functions and the effects of spatial and phase-space refinement.

Appendix A. A Brief Review of the Reconstruction Operators. In this appendix, we review the x -directional reconstruction operators $\mathcal{R}_{j+\frac{1}{2},k}^{(r),\pm}$ at the interface $(x_{j+\frac{1}{2}}, y_k)$. The y -directional reconstruction operators $\mathcal{R}_{j,k+\frac{1}{2}}^{(r),\pm}$ are constructed analogously, and their details are omitted for the sake of brevity. For further details, we refer the reader to [22].

A.1. Local Characteristic Reconstruction. We first define $\hat{A}_{j+\frac{1}{2},k} := A(\hat{U}_{j+\frac{1}{2},k})$, where $A(U) := \frac{\partial \mathbf{F}}{\partial U}$ is the x -directional flux Jacobian, and compute the matrices $R_{j+\frac{1}{2},k}$ and $R_{j+\frac{1}{2},k}^{-1}$ such that $R_{j+\frac{1}{2},k}^{-1} \hat{A}_{j+\frac{1}{2},k} R_{j+\frac{1}{2},k}$ is diagonal. Here, $\hat{U}_{j+\frac{1}{2},k}$ is an average of $U_{j,k}$ and $U_{j+1,k}$, which is obtained by averaging the primitive variables:

$$\hat{\rho}_{j+\frac{1}{2},k} = \frac{\rho_{j,k} + \rho_{j+1,k}}{2}, \quad \hat{u}_{j+\frac{1}{2},k} = \frac{u_{j,k} + u_{j+1,k}}{2}, \quad \hat{v}_{j+\frac{1}{2},k} = \frac{v_{j,k} + v_{j+1,k}}{2}, \quad \hat{p}_{j+\frac{1}{2},k} = \frac{p_{j,k} + p_{j+1,k}}{2}.$$

The split fluxes on the reconstruction stencil are projected onto this single characteristic basis according to

$$\mathbf{Q}_{\ell,k}^{x,\pm} = R_{j+\frac{1}{2},k}^{-1} \mathbf{F}_{\ell,k}^{\pm}, \quad \ell \in \mathcal{S}_{j+\frac{1}{2}}^{(r)}.$$

The scalar left- and right-biased reconstruction operators described below are then applied to each characteristic component to obtain

$$\hat{\mathbf{Q}}_{j+\frac{1}{2},k}^{x,+} = \mathcal{R}_{j+\frac{1}{2}}^{(r),+} \left(\left\{ \mathbf{Q}_{\ell,k}^{x,+} \right\}_{\ell \in \mathcal{S}_{j+\frac{1}{2}}^{(r)}} \right), \quad \hat{\mathbf{Q}}_{j+\frac{1}{2},k}^{x,-} = \mathcal{R}_{j+\frac{1}{2}}^{(r),-} \left(\left\{ \mathbf{Q}_{\ell,k}^{x,-} \right\}_{\ell \in \mathcal{S}_{j+\frac{1}{2}}^{(r)}} \right).$$

The x -directional numerical flux is then recovered from

$$\mathcal{F}_{j+\frac{1}{2},k}^{(r)} = R_{j+\frac{1}{2},k} \left(\hat{\mathbf{Q}}_{j+\frac{1}{2},k}^{x,+} + \hat{\mathbf{Q}}_{j+\frac{1}{2},k}^{x,-} \right).$$

The y -directional numerical fluxes $\mathcal{G}_{j,k+\frac{1}{2}}^{(r)}$ are obtained analogously using the Jacobian $B(U) := \frac{\partial \mathbf{G}}{\partial U}$ and its corresponding left and right eigenvector matrices. In the following subsections, the scalar formulas are applied separately to every characteristic component, and the k -index is suppressed for simplicity.

Remark A.1. For detailed explanations of how the matrices $R_{j+\frac{1}{2},k}$ and $R_{j+\frac{1}{2},k}^{-1}$ are computed for the 2-D Euler equations of gas dynamics, we refer the reader to [1, Appendix C].

A.2. First-Order Reconstruction. For $r = 1$, the reconstruction operators are

$$\mathcal{R}_{j+\frac{1}{2}}^{(1),+}(\{q_\ell^+\}) = q_j^+, \quad \mathcal{R}_{j+\frac{1}{2}}^{(1),-}(\{q_\ell^-\}) = q_{j+1}^-.$$

Their sum gives a first-order LLF-type numerical flux. For the classical schemes, this is the standard LLF flux, whereas for the Young-measure scheme, the physical flux values are replaced by their Young-measure averages.

A.3. Second-Order Reconstruction. For $r = 2$, the one-sided point values are reconstructed as

$$\mathcal{R}_{j+\frac{1}{2}}^{(2),+}(\{q_\ell^+\}) = q_j^+ + \frac{\Delta x}{2}(q_j^+)_x, \quad \mathcal{R}_{j+\frac{1}{2}}^{(2),-}(\{q_\ell^-\}) = q_{j+1}^- - \frac{\Delta x}{2}(q_{j+1}^-)_x,$$

where the slopes $(q_j^\pm)_x$ are computed using a nonlinear limiter to ensure the non-oscillatory property of the reconstruction. In this paper, we use the generalized minmod limiter (see, e.g., [24, 27, 31]):

$$(A.1) \quad (q_j^\pm)_x = \text{minmod} \left(\theta \frac{q_j^\pm - q_{j-1}^\pm}{\Delta x}, \frac{q_{j+1}^\pm - q_{j-1}^\pm}{2\Delta x}, \theta \frac{q_{j+1}^\pm - q_j^\pm}{\Delta x} \right).$$

Here, the minmod function is defined by

$$\text{minmod}(z_1, z_2, \dots) := \begin{cases} \min_\ell \{z_\ell\}, & z_\ell > 0 \quad \forall \ell, \\ \max_\ell \{z_\ell\}, & z_\ell < 0 \quad \forall \ell, \\ 0, & \text{otherwise.} \end{cases}$$

The parameter θ in (A.1) controls the amount of numerical diffusion in the resulting scheme. In general, larger values of θ lead to sharper but potentially more oscillatory reconstructions. In the numerical example reported in §4, we take $\theta = 2$.

A.4. High-Order WENO Reconstructions. The third-, fifth-, seventh-, and ninth-order reconstruction operators are described simultaneously by setting $r = 2s - 1$, with $s = 2, 3, 4, 5$. Following the WENO construction, the reconstructed interface values are convex combinations of candidate values:

$$(A.2) \quad \mathcal{R}_{j+\frac{1}{2}}^{(2s-1),+}(\{q_\ell^+\}) = \sum_{m=0}^{s-1} \omega_m^+ p_{m,j+\frac{1}{2}}^+, \quad \mathcal{R}_{j+\frac{1}{2}}^{(2s-1),-}(\{q_\ell^-\}) = \sum_{m=0}^{s-1} \omega_m^- p_{m,j+\frac{1}{2}}^-.$$

The nonlinear weights are given by $\alpha_m^\pm = \frac{d_m^{(s)}}{(\varepsilon_0 + \beta_m^\pm)^2}$, where $\omega_m^\pm = \alpha_m^\pm / \sum_{\ell=0}^{s-1} \alpha_\ell^\pm$, and $\varepsilon_0 > 0$ is a fixed small parameter. In this paper, we take $\varepsilon_0 = 10^{-12}$ in all of the schemes. The optimal weights are

$$d_m^{(s)} = \frac{\binom{s-1}{m} \binom{s}{m}}{\binom{2s-1}{s}}, \quad m = 0, \dots, s-1.$$

For the fifth-, seventh-, and ninth-order reconstructions, the smoothness indicators are computed by

$$(A.3) \quad \beta_m^+ = \sum_{\nu=1}^{s-1} \Delta x^{2\nu-1} \int_{I_j} \left[\frac{d^\nu}{dx^\nu} p_m^+(x) \right]^2 dx, \quad \beta_m^- = \sum_{\nu=1}^{s-1} \Delta x^{2\nu-1} \int_{I_{j+1}} \left[\frac{d^\nu}{dx^\nu} p_m^-(x) \right]^2 dx,$$

where $I_j = [x_{j-1/2}, x_{j+1/2}]$, and the polynomials $p_m^\pm(x)$ have degree at most $s-1$ and satisfy

$$\frac{1}{\Delta x} \int_{x_{\ell-1/2}}^{x_{\ell+1/2}} p_m^\pm(x) dx = q_\ell^\pm$$

for every index ℓ in the corresponding candidate stencil. This is the standard auxiliary cell-average interpretation used to construct finite-difference WENO numerical fluxes from the pointwise split-flux data $q_\ell^\pm$. We refer to [19] for the explicit forms of the smoothness indicators used in the fifth-, seventh-, and ninth-order schemes. For the third-order scheme, the indicators in (A.3) are replaced by the new smoothness indicators introduced in [9, Appendix A] to achieve higher resolution, while the remaining reconstruction formulas are unchanged.

Finally, the interface values in (A.2) are given by

$$p_{m,j+\frac{1}{2}}^+ = \sum_{\ell=0}^{s-1} C_{m,\ell}^{[s]} q_{j-s+1+m+\ell}^+, \quad p_{m,j+\frac{1}{2}}^- = \sum_{\ell=0}^{s-1} C_{m,\ell}^{[s]} q_{j+s-m-\ell}^-,$$

where the coefficient matrices are given by

$$C^{[2]} = \begin{pmatrix} \frac{1}{2} & \frac{3}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad C^{[3]} = \begin{pmatrix} \frac{1}{3} & -\frac{7}{6} & \frac{11}{6} \\ -\frac{1}{6} & \frac{5}{6} & \frac{1}{3} \\ \frac{1}{3} & \frac{5}{6} & -\frac{1}{6} \end{pmatrix}, \quad C^{[4]} = \begin{pmatrix} -\frac{1}{4} & \frac{13}{12} & -\frac{23}{12} & \frac{25}{12} \\ \frac{1}{12} & -\frac{5}{12} & \frac{13}{12} & \frac{1}{4} \\ -\frac{1}{12} & \frac{7}{12} & \frac{7}{12} & -\frac{1}{12} \\ \frac{1}{4} & \frac{13}{12} & -\frac{5}{12} & \frac{1}{12} \end{pmatrix},$$

and

$$C^{[5]} = \begin{pmatrix} \frac{1}{5} & -\frac{21}{20} & \frac{137}{60} & -\frac{163}{60} & \frac{137}{60} \\ -\frac{1}{20} & \frac{17}{60} & -\frac{43}{60} & \frac{77}{60} & \frac{1}{5} \\ \frac{1}{30} & -\frac{13}{60} & \frac{47}{60} & \frac{9}{20} & -\frac{1}{20} \\ -\frac{1}{20} & \frac{9}{20} & \frac{47}{60} & -\frac{13}{60} & \frac{1}{30} \\ \frac{1}{5} & \frac{77}{60} & -\frac{43}{60} & \frac{17}{60} & -\frac{1}{20} \end{pmatrix}.$$

Appendix B. Entropy-Based Local Characteristic Decomposition. In this appendix, we briefly describe the entropy-based LCD; see, e.g., [6]. The reconstruction procedure is the same as that presented in Appendix A, except that the interface state $\widehat{U}_{j+\frac{1}{2},k}$ used to construct the matrix $\widehat{A}_{j+\frac{1}{2},k} = A(\widehat{U}_{j+\frac{1}{2},k})$ is selected according to the physical entropy, determined by the following algorithm.

Algorithm B.1 Entropy-based average interface state in the x -direction.

```

1: Set  $S_1 := S(\rho_{j,k}, p_{j,k})$ ,  $S_2 := S(\rho_{j+1,k}, p_{j+1,k})$ ,  $S_3 := S(\rho_{j,k}, p_{j+1,k})$ ,  $S_4 := S(\rho_{j+1,k}, p_{j,k})$ 
2: if  $S_1 = \max\{S_1, S_2, S_3, S_4\}$  then
3:   set  $\widehat{\rho}_{j+\frac{1}{2},k} := \rho_{j,k}$ ,  $\widehat{u}_{j+\frac{1}{2},k} := u_{j,k}$ ,  $\widehat{v}_{j+\frac{1}{2},k} := v_{j,k}$ ,  $\widehat{p}_{j+\frac{1}{2},k} := p_{j,k}$ 
4: else if  $S_2 = \max\{S_1, S_2, S_3, S_4\}$  then
5:   set  $\widehat{\rho}_{j+\frac{1}{2},k} := \rho_{j+1,k}$ ,  $\widehat{u}_{j+\frac{1}{2},k} := u_{j+1,k}$ ,  $\widehat{v}_{j+\frac{1}{2},k} := v_{j+1,k}$ ,  $\widehat{p}_{j+\frac{1}{2},k} := p_{j+1,k}$ 
6: else if  $S_3 = \max\{S_1, S_2, S_3, S_4\}$  then
7:   set  $\widehat{\rho}_{j+\frac{1}{2},k} := \rho_{j,k}$ ,  $\widehat{u}_{j+\frac{1}{2},k} := (u_{j,k} + u_{j+1,k})/2$ ,  $\widehat{v}_{j+\frac{1}{2},k} := (v_{j,k} + v_{j+1,k})/2$ ,  $\widehat{p}_{j+\frac{1}{2},k} := p_{j+1,k}$ 
8: else
9:   set  $\widehat{\rho}_{j+\frac{1}{2},k} := \rho_{j+1,k}$ ,  $\widehat{u}_{j+\frac{1}{2},k} := (u_{j,k} + u_{j+1,k})/2$ ,  $\widehat{v}_{j+\frac{1}{2},k} := (v_{j,k} + v_{j+1,k})/2$ ,  $\widehat{p}_{j+\frac{1}{2},k} := p_{j,k}$ 
10: end if

```

The remaining reconstruction steps are exactly the same as those described in Appendix A. The entropy-based interface state in the y -direction is constructed analogously, and its details are omitted for the sake of brevity.

REFERENCES

- [1] A. CHERTOCK, S. CHU, M. HERTY, A. KURGANOV, AND M. LUKÁČOVÁ-MEDVIĐOVÁ, *Local characteristic decomposition based central-upwind scheme*, J. Comput. Phys., 473 (2023). Paper No. 111718.
- [2] A. CHERTOCK, S. CHU, AND A. KURGANOV, *Adaptive high-order A-WENO schemes based on a new local smoothness indicator*, East Asian J. Appl. Math., 13 (2023), pp. 576–609.
- [3] A. CHERTOCK, M. HERTY, A. S. ISKHAPOV, A. ISKHAPOVA, A. KURGANOV, AND M. LUKÁČOVÁ-MEDVIĐOVÁ, *Numerical study of random Kelvin-Helmholtz instability*, Commun. Comput. Phys., 40 (2026), pp. 1394–1415.

- [4] E. CHIODAROLI, C. DE LELLIS, AND O. KREML, *Global ill-posedness of the isentropic system of gas dynamics*, Comm. Pure Appl. Math., 68 (2015), pp. 1157–1190.
- [5] S. CHU AND M. HERTY, *High-Order Schemes for Hyperbolic Conservation Laws Using Young Measures*, J. Comput. Phys., 568 (2027). Paper No. 115375.
- [6] S. CHU, M. HERTY, AND A. KURGANOV, *Entropy-Based Local Characteristic Decomposition*. Submitted. Preprint available at <https://arxiv.org/pdf/2609.19838>.
- [7] S. CHU, M. HERTY, A. KURGANOV, M. LUKÁČOVÁ-MEDVIĐOVÁ, AND C. YU, *Numerical Study of Dissipative Weak Solutions for the Euler Equations of Gas Dynamics*, to appear in Pure Appl. Funct. Anal. Preprint available at <https://arxiv.org/abs/2601.17452>.
- [8] S. CHU, M. HERTY, M. LUKÁČOVÁ-MEDVIĐOVÁ, AND Y. ZHOU, *Solving random hyperbolic conservation laws using linear programming*, SIAM J. Sci. Comput., 48 (2026), pp. A1184–A1205.
- [9] S. CHU, M. HERTY, AND E. F. TORO, *High-order flux splitting schemes for the Euler equations of gas dynamics*, Comput. & Fluids, 300 (2025). Paper No. 106738.
- [10] C. DE LELLIS AND L. SZÉKELYHIDI, JR., *On admissibility criteria for weak solutions of the Euler equations*, Arch. Ration. Mech. Anal., 195 (2010), pp. 225–260.
- [11] R. J. DiPERNA, *Measure-valued solutions to conservation laws*, Arch. Ration. Mech. Anal., 88 (1985), pp. 223–270.
- [12] E. FEIREISL, C. KLINGENBERG, AND S. MARKFELDER, *On the density of “wild” initial data for the compressible Euler system*, Calc. Var. Partial Differential Equations, 59 (2020). Paper No. 152.
- [13] E. FEIREISL, M. LUKÁČOVÁ-MEDVIĐOVÁ, B. SHE, AND Y. WANG, *Computing oscillatory solutions of the Euler system via $\mathcal{K}$ -convergence*, Math. Models Methods Appl. Sci., 31 (2021), pp. 537–576.
- [14] E. FEIREISL AND M. LUKÁČOVÁ-MEDVIĐOVÁ, *Well-posedness of the Euler system of gas dynamics*. Submitted. Preprint available at <https://arxiv.org/abs/2512.18267>.
- [15] E. FEIREISL, M. LUKÁČOVÁ-MEDVIĐOVÁ, AND H. MIZEROVÁ, *A finite volume scheme for the Euler system inspired by the two velocities approach*, Numer. Math., 144 (2020), pp. 89–132.
- [16] E. FEIREISL, M. LUKÁČOVÁ-MEDVIĐOVÁ, H. MIZEROVÁ, AND B. SHE, *Numerical Analysis of Compressible Fluid Flows*, vol. 20 of MS&A. Modeling, Simulation and Applications, Springer, Cham, 2021.
- [17] E. FEIREISL, M. LUKÁČOVÁ-MEDVIĐOVÁ, AND C. YU, *Oscillatory approximations and maximum entropy principle for the Euler system of gas dynamics*. To appear in J. Eur. Math. Soc. Preprint available at <https://arxiv.org/abs/2505.02070>.
- [18] U. S. FJORDHOLM, S. MISHRA, AND E. TADMOR, *On the computation of measure-valued solutions*, Acta Numer., 25 (2016), pp. 567–679.
- [19] Z. GAO, L.-L. FANG, B.-S. WANG, Y. WANG, AND W. S. DON, *Seventh and ninth orders characteristic-wise alternative WENO finite difference schemes for hyperbolic conservation laws*, Comput. & Fluids, 202 (2020). Paper No. 104519.
- [20] S. GOTTLIEB, D. KETCHESON, AND C.-W. SHU, *Strong stability preserving Runge-Kutta and multistep time discretizations*, World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ, 2011.
- [21] S. GOTTLIEB, C.-W. SHU, AND E. TADMOR, *Strong stability-preserving high-order time discretization methods*, SIAM Rev., 43 (2001), pp. 89–112.
- [22] G.-S. JIANG AND C.-W. SHU, *Efficient implementation of weighted ENO schemes*, J. Comput. Phys., 126 (1996), pp. 202–228.
- [23] Y. JIANG, C.-W. SHU, AND M. ZHANG, *An alternative formulation of finite difference weighted ENO schemes with Lax-Wendroff time discretization for conservation laws*, SIAM J. Sci. Comput., 35 (2013), pp. A1137–A1160.
- [24] K.-A. LIE AND S. NOELLE, *On the artificial compression method for second-order nonoscillatory central difference schemes for systems of conservation laws*, SIAM J. Sci. Comput., 24 (2003), pp. 1157–1174.
- [25] H. LIU, *A numerical study of the performance of alternative weighted ENO methods based on various numerical fluxes for conservation law*, Appl. Math. Comput., 296 (2017), pp. 182–197.
- [26] M. LUKÁČOVÁ-MEDVIĐOVÁ, B. SHE, AND Y. YUAN, *What is a limit of structure-preserving numerical methods for compressible flows?*, in Numerical Mathematics and Advanced Applications ENUMATH 2023, Volume 1, A. Sequeira, A. Silvestre, S. S. Valtchev, and J. Janela, eds., vol. 153 of Lecture Notes in Computational Science and Engineering, Springer, Cham, 2025, pp. 17–33.
- [27] H. NESSYAHU AND E. TADMOR, *Nonoscillatory central differencing for hyperbolic conservation laws*, J. Comput. Phys., 87 (1990), pp. 408–463.
- [28] T. NONOMURA AND K. FUJII, *Characteristic finite-difference WENO scheme for multicomponent compressible fluid analysis: overestimated quasi-conservative formulation maintaining equilibriums of velocity, pressure, and temperature*, J. Comput. Phys., 340 (2017), pp. 358–388.
- [29] J. QIU AND C.-W. SHU, *On the construction, comparison, and local characteristic decomposition for high-order central WENO schemes*, J. Comput. Phys., 183 (2002), pp. 187–209.
- [30] C.-W. SHU, *Essentially non-oscillatory and weighted essentially non-oscillatory schemes*, Acta Numer., 29 (2020), pp. 701–762.
- [31] P. K. SWEBY, *High resolution schemes using flux limiters for hyperbolic conservation laws*, SIAM J. Numer. Anal., 21 (1984), pp. 995–1011.